

-Kabir Sinha

-E23CSEU1343

CSET419 – Introduction to Generative AI

Lab 1: Synthetic Image Dataset Generation using Stable Diffusion

```
[1]
✓ 5s
import torch
print("GPU Available:", torch.cuda.is_available())

GPU Available: True

[2]
✓ 6s
!pip install diffusers transformers accelerate safetensors --quiet

[3]
✓ 17s
import torch
from diffusers import StableDiffusionPipeline
import os
import matplotlib.pyplot as plt

Flax classes are deprecated and will be removed in Diffusers v1.0.0. We recommend migrating to PyTorch classes or pinning your version of Diffusers.
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```

```
[4]
✓ 43s
model_id = "runwayml/stable-diffusion-v1-5"

pipe = StableDiffusionPipeline.from_pretrained(
    model_id,
    torch_dtype=torch.float16,
    safety_checker=None
)

pipe = pipe.to("cuda")
pipe.enable_attention_slicing()

print("Model loaded successfully")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your notebook.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
model_index.json: 100% 541/541 [00:00<00:00, 60.8kB/s]
Download complete: 4.27G/7 [00:33<00:00, 138MB/s]
Fetching 13 files: 100% 13/13 [00:33<00:00, 3.49s/file]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading pipeline components...: 100% 6/6 [00:07<00:00, 1.01s/file]
Loading weights: 100% 196/196 [00:01<00:00, 209.67kB/s, Materializing param=text_model.final_layer_norm.weight]
CLIPTextModel LOAD REPORT from: /root/.cache/huggingface/hub/models--runwayml--stable-diffusion-v1-5/snapshots/451f4fe16113bff5a5d2269ed5ad43b0592e9a14/text_encoder
Key | Status |
-----+-----+
text_model.embeddings.position_ids | UNEXPECTED |
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
You have disabled the safety checker for <class 'diffusers.pipelines.stable_diffusion.pipeline_stable_diffusion.StableDiffusionPipeline'> by passing `safety_checker=None`. En
Model loaded successfully
```

```
[5]
✓ 0s
prompts = [
    "A futuristic smart city at night with neon lights, ultra realistic, 4k",
    "A peaceful mountain lake during sunset, cinematic lighting",
    "A cute astronaut cat floating in space, digital art",
    "A vintage steam train crossing a stone bridge in autumn",
    "A humanoid robot playing chess in a quiet park, highly detailed"
]

[6]
✓ 0s
output_folder = "synthetic_dataset"
os.makedirs(output_folder, exist_ok=True)

[7]
✓ 37s
generated_images = []

for i, prompt in enumerate(prompts):
    print(f"Generating image {i+1} ...")

    image = pipe(
        prompt,
        num_inference_steps=30,
        guidance_scale=7.5
    ).images[0]

    image.save(f"{output_folder}/sample_{i+1}.png")
    generated_images.append(image)

print("Dataset generated successfully")
```

Generating image 1 ...

100% 30/30 [00:08<00:00, 5.04it/s]

Generating image 2 ...

100% 30/30 [00:06<00:00, 5.02it/s]

Generating image 3 ...

100% 30/30 [00:07<00:00, 4.91it/s]

Generating image 4 ...

100% 30/30 [00:06<00:00, 4.94it/s]

Generating image 5 ...

100% 30/30 [00:06<00:00, 4.82it/s]

Dataset generated successfully

[8]
✓ 0s
plt.figure(figsize=(15,5))
for i, img in enumerate(generated_images):
 plt.subplot(1,5,i+1)
 plt.imshow(img)
 plt.axis("off")
plt.show()



```
[9]
✓ 0s
!zip -r synthetic_dataset.zip synthetic_dataset

adding: synthetic_dataset/ (stored 0%)
adding: synthetic_dataset/sample_1.png (deflated 0%)
adding: synthetic_dataset/sample_3.png (deflated 0%)
adding: synthetic_dataset/sample_4.png (deflated 0%)
adding: synthetic_dataset/sample_2.png (deflated 0%)
adding: synthetic_dataset/sample_5.png (deflated 0%)
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    "A cute astronaut cat floating in space, digital art",
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    "A humanoid robot playing chess in a quiet park, highly detailed"
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